



Wrocław
University
of Science
and Technology

Software System Development (Level 2) (Lecture)

Project costs estimation

Marek Krótkiewicz

`Marek.Krotkiewicz@pwr.edu.pl`

Department of Applied Informatics
Faculty of Information and Communication Technology
Wrocław University of Science and Technology

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Consultations

Consultations

Consultations: Summer 2026

- Tuesday:

1:00 p.m. – **3:00** p.m. (13:00 – 15:00)

Building: **D2**, Room: **302b**

- Friday:

1:00 p.m. – **3:00** p.m. (13:00 – 15:00)

Zoom platform

The consultations during my office hours are by appointment only. The e-mail with the date and hour of choice should be send at least **2** days in advance with the specific topic of the consultations.

e-mail: Marek.Krotkiewicz@pwr.edu.pl



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Final test

Final test

Exam: Summer 2026

- I term:

Date: **29 June** 2026

Time: **3:15** p.m. – 3:40 p.m. (15:15 – 15:40)

- II term:

Date: **6 July** 2026

Time: **3:15** p.m. – 3:40 p.m. (15:15 – 15:40)

Rules:

- The exam will take place online.
- It will be possible to join the exam only within the first 5 minutes of its start time.
- To take the exam, you need access to the Internet, a computer and an installed web browser.
- No additional dates are planned.
- The exam result is expressed as a percentage and constitutes an element of the assessment for a group of courses, in accordance with the rules presented below.

Agenda

Consultations

Final test

Introduction

An Overview of Cost Estimation

Principles of Cost Estimation

Process for Cost Estimation

Challenges of Cost Estimation

Project Cost Estimation Techniques

Elements of cost estimation in project management

Cost Control



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Introduction



Introduction

Good **cost estimation** is essential for keeping a project under budget. Many costs can appear over the life cycle of a project, and an accurate estimation method can be the difference between a successful plan and a failed one.

Cost estimation is the process that takes those factors into account, and calculates a budget that meets the financial commitment necessary for a successful project.



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An Overview of Cost Estimation

An Overview of Cost Estimation

Basic features of a reliable cost estimate:

- Clear identification of tasks
- Broad participation in preparing estimates
- Availability of valid data
- Standardised structures for the estimates
- Provisions for program uncertainties
- Recognition of inflation
- Excluded costs
- Independent reviews of estimates
- Revision of estimates for significant program changes



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Principles of Cost Estimation

Principles of Cost Estimation

- 1 Cost estimation is used to predict the quantity, cost and price of the resources required by the scope of a project.
- 2 Cost estimation is needed to provide decision-makers with the means to make investment decisions, choose between alternatives and to set up the budget during the front end of projects.
- 3 Estimating is done by breaking down the total scope of a project in manageable parts, to which resources can be assigned and cost.
- 4 A cost estimate is more than a list of costs. It also includes a detailed Basis of Estimate (BOE) report that describes the assumptions, inclusions, exclusions, accuracy and other aspects that are needed to interpret the total project cost.



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Process for Cost Estimation

Process for Cost Estimation

- 1 Define Estimate's Purpose
- 2 Develop Estimating Plan
- 3 Define Characteristics
- 4 Determine Estimating Approach
- 5 Identify Rule and Assumptions
- 6 Obtain Data
- 7 Develop Point Estimate
- 8 Conduct Sensitivity Analysis
- 9 Conduct Risk and Uncertainty Analysis
- 10 Document the Estimate
- 11 Present Estimate to Management
- 12 Update Estimate



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Challenges of Cost Estimation



Challenges of Cost Estimation

There are many factors that are uncertain when cost estimating.

- Planning horizon is further out.
- The project is not like prior ones.
- The team: the level of skill and experience



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Project Cost Estimation Techniques



Project Cost Estimation Techniques

- Analogous Estimating
- Statistical Modeling
- Bottom-Up Estimating
- Three-point Estimate
- Reserve Analysis
- Cost of Quality
- Dynamic Tools



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Elements of cost estimation in project management



Elements of cost estimation in project management

Key cost **types**:

- direct costs,
- indirect costs.

Typical cost **elements**:

- labor,
- materials and equipment,
- facilities,
- vendors,
- risk.



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Cost Control

Cost Control

Cost Control is concerned with measuring variances from the cost baseline and taking effective corrective action to achieve minimum cost overruns.

You continuously have to be aware of these elements:

- Know what has to be done
- Know what has been done
- Know how actual performance compares with performance norms
- Know what remains to be done
- Identify and implement corrective actions
- Check the results of corrective actions

Thank you for your attention